

Lecture 10: The Chain Rule

GOAL: Be able to differentiate functions of the form $f(g(x))$.

Summary:

$$\ln(x^2) \quad e^{3x}$$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\frac{d}{dx}[f(g(x))] = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$[g(x)^p]' = p[g(x)]^{p-1} \cdot g'(x) \quad [b^x]' = b^x \cdot \ln b$$

Theorem (Chain Rule) Let f and g be diff fcn

Format I: $\frac{d}{dx}(f \circ g) = \frac{df}{dg} \cdot \frac{dg}{dx}$
 $f(g(x))$

Format II: $[f(g(x))]' = f'(g(x)) \cdot g'(x)$

Easy (but incorrect) reason why

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} \left(\frac{\Delta g}{\Delta g} \right) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta g} \frac{\Delta g}{\Delta x} = \frac{df}{dg} \frac{dg}{dx}$$

Example 1 $\frac{d}{dx}(x^{\frac{3}{2}}) = \frac{3}{2} x^{\frac{1}{2}}$ | $[x^{\frac{1}{2}}]^3$ $x \mapsto x^{\frac{1}{2}} \mapsto (x^{\frac{1}{2}})^3$

$$g(x) = x^{\frac{1}{2}}$$

$$f(g) = g^3$$

Format I: $\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx} = \frac{d}{dg}(g^3) \cdot \frac{d}{dx}(x^{\frac{1}{2}})$
 $= 3g^2 \cdot \frac{1}{2} x^{-\frac{1}{2}} = 3[x^{\frac{1}{2}}]^2 \left(\frac{1}{2} x^{-\frac{1}{2}} \right)$
 $= \frac{3}{2} x^{\frac{1}{2}}$

$$\begin{aligned}\text{Format II: } [(x^{\frac{1}{2}})^3]' &= f'(g(x)) \cdot g'(x) \\ &= \underbrace{[g^3]'}_{\text{w.r.t. } g} \cdot \underbrace{[x^{\frac{1}{2}}]'}_{\text{w.r.t. } x} \\ &= 3g^2 \cdot \frac{1}{2}x^{-\frac{1}{2}} = \dots = \frac{3}{2}x^{\frac{1}{2}}\end{aligned}$$

NOTE: This gives us a way of showing the power rule if the exponent is rational

$$x^{\frac{m}{n}} = (x^{\frac{1}{n}})^m \quad \begin{matrix} \uparrow \\ \text{Integers} \end{matrix}$$

Ex 2 $[(\sin x)^2]'$ $g(x) = \sin x$
 $f(g) = g^2$

$$\begin{aligned}\text{Format 1: } \frac{df}{dx} &= \frac{df}{dg} \cdot \frac{dg}{dx} = \frac{d}{dg}(g^2) \cdot \frac{d}{dx}(\sin x) \\ &= 2g \cdot \cos x = 2 \sin x \cos x\end{aligned}$$

$$\begin{aligned}\text{Format 2: } [(\sin x)^2]' &= f'(g(x)) \cdot g'(x) \\ &= \underbrace{[g^2]'}_{\text{w.r.t. } g} \cdot \underbrace{[\sin x]'}_{\text{w.r.t. } x} \\ &= 2g \cos x = 2 \sin x \cos x\end{aligned}$$

Ex 3 $y = e^{x^2+x+1}$ $\left[\begin{array}{l} x \mapsto x^2+x+1 \\ \mapsto e^{x^2+x+1} \end{array} \right] \begin{array}{l} g(x) = x^2+x+1 \\ f(g) = f(g) \end{array}$

$$\frac{dy}{dx} = \frac{d}{dg}(e^g) \cdot \frac{d}{dx}(x^2+x+1) = e^g \cdot (2x+1) \\ = (2x+1) e^{x^2+x+1}$$

NOTE: It is common to skip the step of writing f as a fn of g

$$y' = [e^{x^2+x+1}]' = \underbrace{[e^{x^2+x+1}]'}_{\substack{\text{w.r.t.} \\ "x^2+x+1"}} \cdot \underbrace{[x^2+x+1]'}_{\text{w.r.t. } x} \\ = e^{x^2+x+1} (2x+1)$$

Example 4 $w(x) = \tan(\cos(x^2))$

$$w'(x) = \sec^2(\cos(x^2)) \cdot [\cos(x^2)]' \\ = \sec^2(\cos(x^2)) \cdot -\sin(x^2) \cdot \underbrace{[x^2]'}_{2x} \\ = -2x \sin(x^2) \sec^2(\cos(x^2))$$

Applications of The Chain Rule

Ex 5 Differentiate $y = \boxed{x^3-1}^{100}$ $g(x) = x^3-1$
 $f(-) = (-)^{100}$

$$y' = 100(x^3-1)^{99} \cdot [x^3-1]' \\ = 100(x^3-1)^{99} \cdot 3x^2 = 300x^2(x^3-1)^{99}$$

Theorem (Generalized Power Rule)

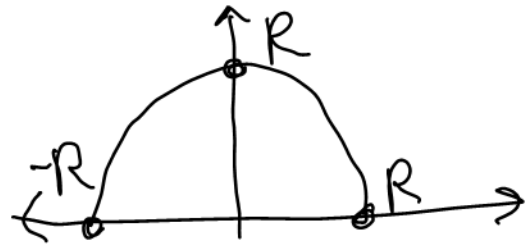
Let $g(x)$ is diff and p is any real number

$$[(g(x))^p]' = p(g(x))^{p-1} \cdot g'(x)$$

When $g(x)=x$, this is the usual power rule.

Ex 6 Let $R > 0$. The semicircle of radius R is given by

$$y = \sqrt{R^2 - x^2}$$



Find y' :

$$y' = [\sqrt{R^2 - x^2}]' = \left[\underbrace{(R^2 - x^2)}_g^{\frac{1}{2}} \right]'$$

$$f(-) = (-)^{\frac{1}{2}}$$

$$= \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} \cdot [R^2 - x^2]'$$

$$= \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} (-2x) = \dots = \frac{-x}{\sqrt{R^2 - x^2}}$$

Ex 7 What's $\frac{d}{dx} (2^x)$?

$$[2^x]' = [e^{\ln(2^x)}]' = [e^{x(\ln 2)}]'$$

$$\begin{aligned} g(x) &= (\ln 2)x \\ f(-) &= e^- \end{aligned}$$

$$= e^{x(\ln 2)} \cdot \ln 2 = 2^x \cdot \ln 2$$

In general, if $b > 0$

$$\frac{d}{dx}(b^x) = b^x \cdot \ln b$$

Ex 8 (Damped Pendulum) A pendulum's position is measured from its angle from its resting place

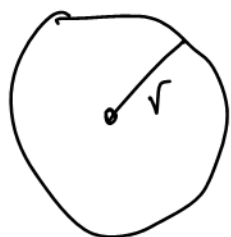


$$s(t) = \left(e^{-t} \cos t \right) \text{ rad}$$

Find $v(t)$

$$\begin{aligned} v(t) = s'(t) &= [e^{-t}]' \cos t + e^{-t} [\cos t]' \\ &= -e^{-t} \cos t + e^{-t} (-\sin t) \\ &= \left(-e^{-t} [\cos t + \sin t] \right) \frac{\text{rad}}{\text{s}} \end{aligned}$$

Ex 9 The radius of a circle is evolving over time. How fast does the area change as a fcn of time.



$$A = \pi r^2$$

$$A(t) = \pi [r(t)]^2$$

$$\begin{aligned} \frac{dA}{dt} &= \pi \cdot 2[r(t)] \cdot \frac{dr}{dt} \\ &= 2\pi r \frac{dr}{dt} \end{aligned}$$

Ex 10 Recall π radians $= 180^\circ \Rightarrow \frac{\pi}{180} \frac{\text{rad}}{1^\circ}$

$$\sin(x^\circ) = \sin\left(\underbrace{\frac{\pi}{180} x}_{\text{radians}}\right)$$

$$\begin{aligned} [\sin(x^\circ)]' &= \left[\sin\left(\frac{\pi}{180} x\right) \right]' = \cos\left(\frac{\pi}{180} x\right) \cdot \frac{\pi}{180} \\ &= \frac{\pi}{180} \cos\left(\frac{\pi}{180} x\right) \\ &= \frac{\pi}{180} \cos(x^\circ) \end{aligned}$$